A published standalone section (Tag `0SE01`)

Definition 1 (Readable typography) (Tag `0SE0101`). A template is readable when its visual hierarchy supports the mathematics without competing with it.

The local references are Section 1 (Tag `0SE01`) and Definition 1 (Tag `0SE0101`).

1 A tagged subsection (Tag `0SE0102`)

Section mode uses a centered page number and no header or footer rules.

A draft environment gallery (Draft tag `0SE02`)

1 Every statement style (Draft tag `0SE0201`)

Slogan *Short independently compiled sections should still share the book's voice.*

Theorem A (Standalone coherence). A standalone section can use the same source as its chapter and book builds.

Definition 2. A chain complex is a sequence (C_n, d_n) with $d_{n-1} \circ d_n = 0$.

Proposition 3 (Cycles modulo boundaries). The quotient $\ker d_n / \operatorname{im} d_{n+1}$ is an abelian group.

Theorem 4. A short exact sequence of complexes induces a long exact sequence in homology.

Lemma 5 (Connecting map). The connecting morphism is independent of the chosen lift.

Corollary 6. A quasi-isomorphism induces isomorphisms on every homology group.

Conjecture 7. An analogous statement holds in the intended derived enhancement.

Question 8 (Naturality). How does the connecting map vary with a morphism of short exact sequences?

Example 9 (Draft tag `0SE0202`). The complex concentrated in degree zero recovers an ordinary module. Its displayed draft tag does not appear in Example 9.

Exercise 10. Write down the connecting map and verify that it is a homomorphism.

Construction 11 (Mapping cone). For $f : C_\bullet \to D_\bullet$, set $\operatorname{Cone}(f)_n = D_n \oplus C_{n-1}$ with the usual differential.

Notation 12. Write $H_n(C_\bullet)$ for the n-th homology group.

Remark 13 (Shared numbering). Every ordinary theorem-like environment participates in one local sequence.

Proof sketch of Theorem 4. Lift a cycle, apply the differential, and descend the resulting boundary.

Step 1 (Choose a lift). Select a representative in the middle complex.

Claim 14. Changing the lift changes the result by a boundary.

Case 1 (The class is zero). Choose a boundary representative and compute directly.

Case 2 (The class is arbitrary). Subtract two lifts and reduce to the first case.

□

References include Theorem A, Theorem 4, Lemma 5, Claim 14, Step 1, and Case 1. Draft structural references such as Section 2 and Subsection 1 omit tags.

2 Mathematics and code (Draft tag `0SE0203`)

Inline mathematics remains calm in prose: $\mathcal{L} \otimes \mathscr{F} \in D^b(X)$ and $\mathfrak{m}_x/\mathfrak{m}_x^2 \cong \mathbb{A}_k^n$. The same notation works in a labeled display:

$$\cdots \longrightarrow C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \longrightarrow \cdots . \tag{1}$$

$$\begin{tikzcd}
0 \arrow[r] & A_\bullet \arrow[r] \arrow[d] & B_\bullet \arrow[r] \arrow[d] & C_\bullet \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & A'_\bullet \arrow[r] & B'_\bullet \arrow[r] & C'_\bullet \arrow[r] & 0
\end{tikzcd}$$

```python
def homology(cycles, boundaries):
    return cycles.quotient(boundaries)
```

Listing 1: A tiny section-level listing

See eq. (1) and Code 1.